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Casevo: A Cognitive Agents and Social Evolution Simulator

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Abstract

In this paper, we introduce a multi-agent simulation framework Casevo (Cognitive Agents and Social Evolution Simulator), that integrates large language models (LLMs) to simulate complex social phenomena and decision-making processes. Casevo is designed as a discrete-event simulator driven by agents with features such as Chain of Thoughts (CoT), Retrieval-Augmented Generation (RAG), and Customizable Memory Mechanism. Casevo enables dynamic social modeling, which can support various scenarios such as social network analysis, public opinion dynamics, and behavior prediction in complex social systems. To demonstrate the effectiveness of Casevo, we utilize one of the U.S. 2020 midterm election TV debates as a simulation example. Our results show that Casevo facilitates more realistic and flexible agent interactions, improving the quality of dynamic social phenomena simulation. This work contributes to the field by providing a robust system for studying large-scale, high-fidelity social behaviors with advanced LLM-driven agents, expanding the capabilities of traditional agent-based modeling (ABM). The open-source code repository address of casevo is `https://github.com/rgCASS/casevo`.

Keywords: Social Simulation, LLM, Agent, MAS, Casevo

1 Introduction

In recent years, with the rapid development of artificial intelligence technology, large language models (LLMs) and agents have shown great potential in various fields. In social science and behavioral research, conducting simulations with agents is one of the commonly used research methods in sociology and communication studies [1]. It can help researchers understand complex social phenomena and human behavior [2]. Traditional social network simulation systems often rely on agents with pre-defined rules and Mathematical Function Models, making it difficult to capture the dynamics and complexity of human behavior [3]. In contrast, agents with LLMs are capable of generating highly realistic natural language texts, simulating human decision-making processes and

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Casevo：认知智能体与社会演化模拟器

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摘要

本文介绍了一种名为 Casevo（认知智能体与社会演化模拟器）的多智能体仿真框架，该框架整合大型语言模型（LLMs），用于模拟复杂社会现象与决策过程。Casevo 被设计为一种由智能体驱动的离散事件仿真器，具备思维链（CoT）、检索增强生成（RAG）及可定制记忆机制等特性。Casevo 支持动态社会建模，可应用于社交网络分析、公众舆论动态演变以及复杂社会系统中的行为预测等多种场景。为验证 Casevo 的有效性，我们以 2020 年美国中期选举电视辩论之一作为仿真案例。实验结果表明，Casevo 能够促成更真实、更灵活的智能体交互，从而提升动态社会现象仿真的质量。本研究通过构建一套基于先进 LLM 驱动智能体的稳健系统，用于探究大规模、高保真社会行为，为相关领域做出了重要贡献，进而拓展了传统基于智能体的建模（ABM）能力。Casevo 的开源代码仓库地址为https://github.com/rgCASS/casevo。

关键词：社会模拟、LLM、智能体、多智能体系统、Casevo

1 引言

近年来，随着人工智能技术的快速发展，大型语言模型（LLMs）与智能体已在多个领域展现出巨大潜力。在社会科学及行为研究领域，利用智能体进行模拟已成为社会学与传播学研究中常用的研究方法之一[1]，这有助于研究人员理解复杂社会现象与人类行为[2]。传统社交网络模拟系统通常依赖于具有预设规则及数学函数模型的智能体，因而难以捕捉人类行为的动态特性与复杂性[3]。相比之下，基于LLMs的智能体能够生成高度逼真的自然语言文本，进而模拟人类决策过程及

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behavioral patterns [4]. Therefore, multi-agent simulation can significantly enhance the realism and effectiveness of social simulation.

In this paper, we introduce Casevo (a Cognitive Agents and Social Evolution Simulator), a multi-agent framework, focusing on the simulation of social interaction and communication based on complex networks. Casevo is built upon the Mesa framework with features for complex and realistic social behaviors and decision-making processes [5]. At the micro level, Casevo supports the construction of highly customized Agents with various features, including Role Insertion, Chain of Thought (CoT), Long-term and Short-term Memory Mechanisms, and so on. These features extend the reasoning capabilities of agents, enabling them to simulate more complex reasoning processes, including event planning and reflection. At the overall level, Casevo supports customized networks for agent interaction, global events, and also supports the parallel optimization of requests to base LLMs.

To demonstrate the advantages and potential of Casevo, one of the U.S. 2020 midterm election TV debates is utilized as a simulation example, in which agents represent voters, and reproduce voter behavior through the election process. The simulation demonstrates the effectiveness of Casevo and shows that it may provide a new methodology and tool for future social studies.

The paper is structured as follows: Section 2 reviews related work. Section 3 is the overview of Casevo. Section 4 details the module implementation. Section 5 shows the concrete implementation of the election simulation and the analysis of the results. Section 6 discusses the future work. Section 7 summarizes the main contributions of the paper. Through this study, we hope to provide a new perspective for social science and behavioral research, promote the development of multi-agent systems, and explore the application potential of LLMs in a wider range of domains.

2 Related Work

This section discusses two key areas most relevant to this research: Multi-Agent Systems and Social Network Analysis. These areas form the core foundation for the development of a social network simulation system.

2.1 Multi-Agent System (MAS)

MAS has emerged as a powerful paradigm for simulating and solving complex problems in a variety of domains, including social simulations, robotics, economics, and environmental management. MAS involves multiple autonomous agents, each of which can make decisions and interact with other agents within a shared environment. The interactions between agents are often governed by predefined rules, but the overall system's behavior can emerge as a result of these interactions. MAS has been particularly useful in modeling scenarios where individual agents have diverse goals, strategies, and levels of autonomy [6].

The integration of LLMs into MAS has opened new possibilities for creating more sophisticated agents capable of engaging in complex decision-making, problem-solving, and negotiations. LLMs, with their ability to generate human-like text and handle nuanced contextual information, enable agents to interact in ways that were previously unattainable. For instance, LLM-Debate [7] introduces the concept of debate, allowing agents to engage with the responses of their peers. When these responses conflict with an agent's judgments, a mental argumentation process unfolds, leading to more refined solutions. GPT-Negotiation [8] illustrates how LLMs can autonomously engage in negotiation tasks, demonstrating both strategic reasoning and adaptive learning, a critical advancement for

行为模式[4]。因此，多智能体仿真可显著提升社会仿真的真实感与有效性。

本文介绍了 Casevo（一种认知智能体与社会演化模拟器）——一个多智能体框架，专注于基于复杂网络的社交互动与通信仿真。Casevo 构建于 Mesa 框架之上，具备模拟复杂且真实的社交行为及决策流程的功能[5]。在微观层面，Casevo 支持构建具有多种特性的高度定制化智能体，包括角色注入、思维链（CoT）、长期与短期记忆机制等。这些特性拓展了智能体的推理能力，使其能够模拟更复杂的推理过程，例如事件规划与反思。在宏观层面，Casevo 支持为智能体交互构建定制化网络、管理全局事件，并支持对基础大语言模型（LLM）的请求进行并行优化。

为展示 Casevo 的优势与潜力，本研究以 2020 年美国中期选举中的一场电视辩论作为模拟案例，其中智能体代表选民，并在选举过程中复现选民的投票行为。该模拟实验验证了 Casevo 的有效性，并表明其可为未来社会科学研究提供一种新的方法论与工具。

本文结构如下：第 2 节综述相关研究工作；第 3 节概述 Casevo 系统；第 4 节详述各模块的实现细节；第 5 节展示选举模拟的具体实现及结果分析；第 6 节探讨未来研究方向；第 7 节总结本文的主要贡献。本研究旨在为社会科学及行为科学研究提供新视角，推动多智能体系统的发展，并探索大语言模型（LLMs）在更广泛领域中的应用潜力。

2 相关工作

本节探讨与本研究最为相关的两个关键领域：多智能体系统与社交网络分析。这两个领域构成了构建社交网络仿真系统的核心基础。

2.1 多智能体系统（MAS）

多智能体系统（MAS）已成为在多个领域——包括社会模拟、机器人技术、经济学及环境管理——中模拟并解决复杂问题的强大范式。MAS 由多个自主智能体组成，每个智能体均能在共享环境中做出决策，并与其他智能体进行交互。智能体间的交互通常遵循预定义的规则，但整个系统的行为却可能由这些交互共同涌现而出。MAS 在建模那些个体智能体具有多样化目标、策略及不同自主性水平的场景时，尤其展现出显著优势[6]。

将大语言模型（LLMs）整合到多智能体系统（MAS）中，为构建更具复杂性、能够参与复杂决策、问题求解及谈判的智能体开辟了新途径。大语言模型凭借其生成类人文本及处理细微上下文信息的能力，使智能体得以以以往无法实现的方式进行交互。例如，LLM-Debate [7]引入了辩论概念，允许智能体与同伴的回应进行互动；当这些回应与智能体的判断相冲突时，便会启动内部论证过程，进而催生更优化的解决方案。GPT-Negotiation [8]则展示了大语言模型如何自主参与谈判任务，既体现战略推理能力，又具备自适应学习能力——这对多智能体系统而言是一项关键性突破。

MAS. ChatEval [9] establishes a role-playing multi-agent referee team that assesses the quality of text generated by LLMs through self-initiated debates. This process achieves a level of excellence comparable to that of human reviewers. Corex [10] incorporates a variety of collaborative paradigms, including Debate, Review and Retrieve modes, which collectively enhance the factuality, fidelity and reliability of the argumentation process. These paradigms encourage task-agnostic approaches that allow LLMs to think outside the box, effectively counteracting hallucinations and leading to more accurate solutions.

The application of MAS in social network simulations has proven valuable in capturing the intricacies of human behavior and social dynamics. Recent work by Zhou et al. (2022) [11] utilized MAS to simulate multi-agent interactions in a multiplayer game environment, using graph theory and system transformation methods to model the interactions. This study underscored the importance of network structures in MAS and showed how agent behavior can be influenced by network topology, which is central to social network modeling. In parallel, advancements in incorporating emotion and behavior-driven agents, as seen in the humanoid agent models discussed by Wang et al. (2022)[12], provide a more realistic simulation of human-like interactions by embedding psychological elements such as emotions and relational closeness into agent behavior. Systems like WarAgent [12] demonstrate how MAS can model large-scale international conflicts, simulating complex alliances and responses based on evolving political and social networks. Similarly, S3 [13] applies MAS to simulate public opinion evolution in social networks, where agents' opinions shift in response to external stimuli and interactions, thus offering insights into collective behavior dynamics.

Recent developments have introduced new frameworks for combining MAS with LLMs, such as Swarm by OpenAI (2023), a lightweight framework designed to facilitate the integration of multi-agent systems with LLMs. This approach makes it easier to deploy complex multi-agent simulations across a variety of domains, streamlining the orchestration of agents and improving scalability. By leveraging LLMs for agent interaction and decision-making, these systems are capable of simulating more sophisticated social processes, such as negotiation, persuasion, and opinion dynamics.

2.2 Social network analysis

Social Network Analysis (SNA) has long been employed as a powerful methodology for studying the structure and dynamics of social networks. SNA traditionally uses graph theory, statistical methods, and machine learning techniques to analyze relationships between nodes and edges within a network, aiming to identify central nodes, community structures, and paths of information flow [14]. For example, Friedkin and Johnsen (1997) [15] [16] explored how social influence processes could be modeled in dynamic networks where the structure of the network itself evolves over time. As Artificial Intelligence (AI) technology continues to develop, social simulation research has also evolved. Vespignani (2012) [17] demonstrated how SNA combined with computational models could capture the spread of rumors, diseases, or even misinformation in a more realistic and actionable way. However, these traditional approaches often rely on static data, which limits their ability to adapt to the dynamic, evolving nature of social networks. In the context of dynamic social behaviors—such as opinion shifts, rumor propagation, or public sentiment—traditional SNA is often insufficient.

Recent advancements have sought to address this limitation by integrating SNA with agent-based modeling (ABM) and multi-agent systems (MAS), where individual agents exhibit dynamic behaviors that evolve over time. This approach allows for the representation of not just the static structure of networks but also the dynamic, often unpredictable interactions among agents. It enables a

ChatEval [9]构建了一个角色扮演式多智能体裁判团队，通过自主发起的辩论来评估大语言模型生成文本的质量；这一过程所达到的卓越水准，可与人工评审团队相媲美。Corex [10]整合了多种协作范式，包括“辩论”、“审阅”与“检索”模式，这些模式共同提升了论证过程的事实性、忠实度与可靠性。这些范式促进了与任务无关的方法应用，使大语言模型能够突破思维定式，有效抑制幻觉现象，进而得出更精准的解决方案。

MAS 在社交网络模拟中的应用已被证明对于捕捉人类行为及社会动态的复杂性具有重要价值。Zhou 等人（2022）[11]的最新研究利用 MAS 在多人游戏环境中模拟多智能体交互，采用图论与系统变换方法对交互过程进行建模。该研究强调了网络结构在 MAS 中的重要性，并展示了智能体行为如何受网络拓扑结构影响——这一特性正是社交网络建模的核心所在。与此同时，随着将情感与行为驱动型智能体整合的技术进步——如 Wang 等人（2022）[12]所探讨的人形智能体模型——通过将情感与关系亲密度等心理要素嵌入智能体行为，已能实现对类人交互更真实的模拟。诸如 WarAgent [12]等系统则展示了 MAS 如何用于模拟大规模国际冲突，基于不断演变的政治与社会网络，模拟复杂的联盟关系及应对策略。类似地，S3 [13]将 MAS 应用于模拟社交网络中的舆论演化过程：智能体的观点会根据外部刺激与交互作用而发生变化，从而为理解集体行为动力学提供重要见解。

近期研究进展引入了将多智能体系统（MAS）与大语言模型（LLM）相结合的新型框架，例如 OpenAI 推出的 Swarm（2023）——这是一个轻量级框架，旨在促进多智能体系统与大语言模型的集成。该方法有助于在多个领域部署复杂的多智能体仿真系统，简化智能体编排流程，并提升系统可扩展性。通过利用大语言模型实现智能体交互与决策，此类系统可模拟更复杂的社会交互过程，例如谈判、说服及观点动态演变。

2.2 社交网络分析

社会网络分析（SNA）长期以来一直被用作研究社会网络结构与动态变化的强大方法。SNA 传统上运用图论、统计方法和机器学习技术，来分析网络中节点与边之间的关系，旨在识别核心节点、社区结构以及信息传播路径[14]。例如， Friedkin 与 Johnsen（1997）[15] [16]探讨了在网络结构随时间演化的动态网络中，如何对社会影响过程进行建模。随着人工智能（AI）技术的不断发展，社会模拟研究也取得了长足进步。Vespignani（2012）[17]证明了将 SNA 与计算模型相结合，可以以更真实、更具操作性的方式捕捉谣言、疾病甚至虚假信息的传播过程。然而，这些传统方法往往依赖于静态数据，这限制了它们适应社会网络动态演变特性的能力。在动态社会行为场景下——例如观点转变、谣言传播或公众情绪演变——传统社会网络分析（SNA）往往难以充分应对。

近期研究致力于通过将 SNA 与基于智能体的建模（ABM）及多智能体系统（MAS）相结合，以应对这一局限性；其中，每个智能体均展现出随时间演化的动态行为。该方法不仅能刻画网络的静态结构，还能捕捉智能体间动态且往往不可预测的交互作用，从而有助于更深入地理解网络

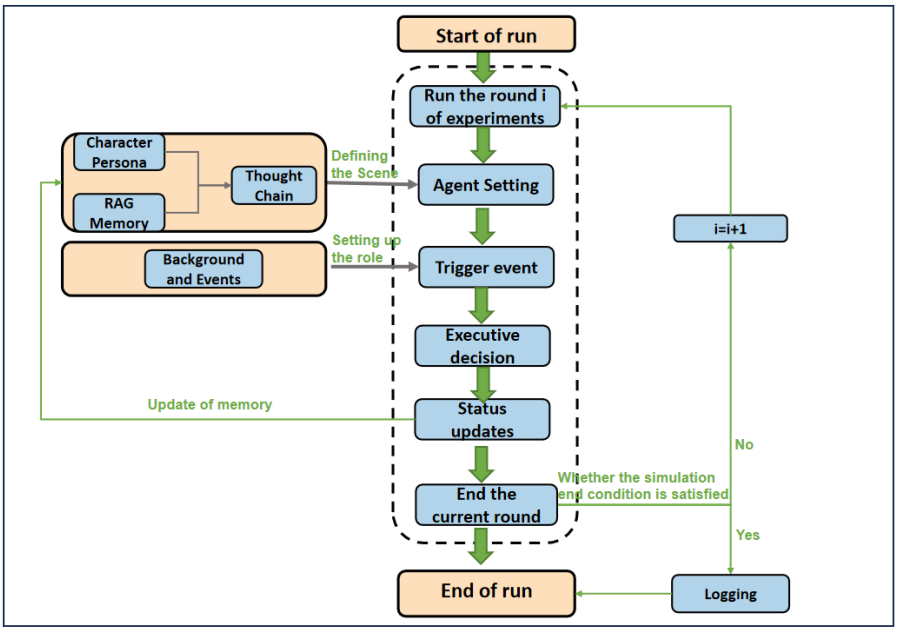


Figure 1: Schematic implementation of Casevo rounds.

more nuanced understanding of how networks function and evolve. For example, in social network simulations, agents' behaviors, such as communication, collaboration, and conflict resolution, can now be modeled dynamically, taking into account both network structure and individual agent behavior [18] [19]. Alvim M.S. and Amorim B. (2023) [20] proposed a polarization model based on confirmation bias, analyzing how users gradually form extreme opinions within social networks. Their study demonstrated that the degree of polarization is influenced by both the network topology and the initial distribution of users' opinions. Similarly, Vilone D. and Polizzi E. (2024) [21] using a multi-agent simulation model, explored opinion misunderstandings and the spiral of silence within various network topologies. They highlighted that highly connected nodes in the network, such as opinion leaders, may play a critical role in shaping information dissemination. Systems like Generative agents [22] leverage agent-based simulations to emulate human behavior in social settings, allowing for the generation of realistic, dynamic interactions that traditional SNA methods could not address. Furthermore, applications such as WarAgent and S3 [12] [13] have shown the potential of these hybrid models in simulating large-scale social phenomena like international conflicts and public opinion evolution, further proving the robustness of combining SNA with agent-based modeling.

3 Overview

Compared to existing open-source MASs, Casevo is specifically designed and built for simulating social interactions on complex networks. Casevo is built upon the Mesa framework [5], designed for discrete-event simulation, using a round-robin updating mechanism to manage the behavior of agents and event scheduling. While the Mesa framework has been widely used in the field of ABM,

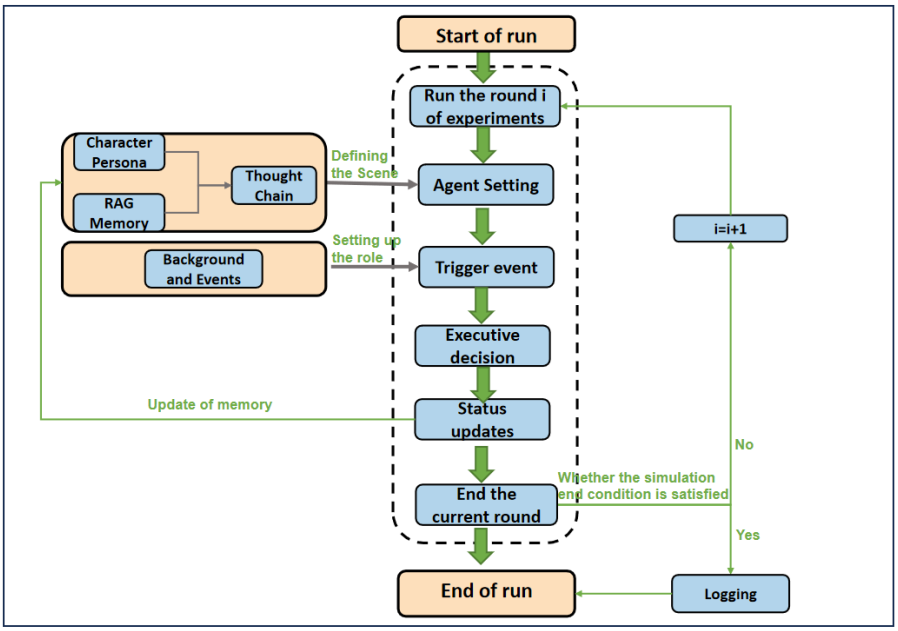


图 1：Casevo 轮次的示意图。

的功能机制与演化规律。例如，在社交网络仿真中，智能体的行为（如沟通、协作与冲突解决）现已可基于网络结构与个体智能体行为，进行动态建模[18] [19]。Alvim M.S. 与 Amorim B.（2023）[20]提出了一种基于确认偏误的极化模型，探究用户如何在社交网络中逐步形成极端观点；其研究证实，极化程度既受网络拓扑结构影响，也受用户初始观点分布的影响。类似地，Vilone D. 与 Polizzi E.(2024)[21]采用多智能体仿真模型，探究了不同网络拓扑结构下的观点误解与沉默螺旋现象。研究强调，网络中高度连接的节点（如意见领袖）可能在塑造信息传播过程中发挥关键作用。基于智能体的系统[22]利用智能体仿真技术模拟人类在社交环境中的行为，从而生成传统社会网络分析（SNA）方法无法实现的逼真、动态交互场景。此外，WarAgent 与 S3 [12] [13]等应用已证实此类混合模型在模拟国际冲突及舆论演变等大规模社会现象方面的潜力，进一步证明了将 SNA 与智能体建模相结合的强大效能。

3 概述

与现有的开源多智能体系统（MAS）相比，Casevo 是专为在复杂网络上模拟社会交互而专门设计并构建的。Casevo 基于为离散事件仿真设计的 Mesa 框架[5]构建，采用轮询更新机制来管理智能体

Casevo extends features as complex semantic reasoning and social network interactions for modeling of complex social phenomena such as election simulation and behavior prediction.

Figure 1 shows a typical example of Casevo's round-based execution. In each round, all agents update their decisions and perform the corresponding behaviors based on the current state, environmental inputs, and interactions with other agents. The turn-by-turn updating ensures the synchronization of system behaviors and the orderly scheduling of events, which is particularly suitable for social dynamic simulation scenarios that need to progress step-by-step.

As shown in Figure 2, Casevo utilizes a modular design manner and integrates the functions of agent behavior generation, memory management, network interaction, and parallel processing to form a highly flexible and scalable system. The system is capable of customizing the simulation scenarios according to the user's needs, including the personality settings of the agent, the network topology, and the intervention of external events. These customization features make Casevo applicable to a wide range of research areas, including social network analysis, the study of public opinion dynamics, the prediction of behavior in complex systems, and the simulation of decision-making processes in collective environments.

The architecture of Casevo is composed of 4 key modules: the Model Module, the Agent Module, the Parallel Optimization Module, and the Network Module. These components work together to support the simulation of complex systems, enabling high-level interactions between agents while efficiently managing large-scale computations. Each module plays a distinct role in managing agent behaviors, decision-making processes, memory recall, and network interactions, ensuring a seamless and efficient simulation flow.

4 Module Implementation

Through the design and integration of four modules, Casevo achieves comprehensive simulation from autonomous decision-making of agents to complex social networks. It provides an efficient and flexible platform for the research and exploration of a wide range of complex social phenomena.

4.1 The Model Module

The Model Module serves to set up the environment where the simulation takes place. It is responsible for scenario settings, event scheduling, interaction channels, prompt templates, and model interfaces. In addition, it handles state logging, ensuring that all changes and actions are recorded throughout the simulation.

- Scenario Settings: Casevo reads the configuration parameters provided by the users, defining the initial environment of the simulation. This includes determining the number of agents, their roles, and other key parameters necessary for the simulation.
- Event Scheduling: The Model Module controls the scheduling of events within the simulation. It ensures that interactions between agents occur on time, adhering to the logical flow defined by the round-based update mechanism.
- Interaction Channels: Agents communicate through well-defined interaction channels, allowing them to share information, influence each other, and adapt their behavior accordingly. These channels replicate real-world communication pathways, such as social media, public debates, or direct conversations.

Casevo 扩展了诸如复杂语义推理与社交网络交互等功能，用于对选举模拟及行为预测等复杂社会现象进行建模。

图1展示了 Casevo 基于轮次的执行过程的典型示例。在每一轮中，所有智能体均根据当前状态、环境输入以及与其他智能体的交互，更新其决策并执行相应行为。这种逐轮更新机制确保了系统行为的同步性以及事件的可控有序调度，尤其适用于需要逐步推进的社会动态仿真场景。
的行为及事件调度。尽管 Mesa 框架已在多智能体建模（ABM）领域广泛应用，但如图2所示，Casevo 采用模块化设计模式，整合了智能体行为生成、内存管理、网络交互及并行处理等功能，从而构建出一个高度灵活且可扩展的系统。该系统可根据用户需求定制仿真场景，包括智能体个性设置、网络拓扑结构以及外部事件的干预。这些定制化特性使 Casevo 能够应用于广泛的研究领域，涵盖社交网络分析、公众舆论动态研究、复杂系统行为预测以及集体环境中的决策过程仿真。

Casevo 的架构由四个核心模块构成：模型模块、智能体模块、并行优化模块及网络模块。这些组件协同工作，支持复杂系统的仿真，既实现智能体间的高层交互，又能高效管理大规模计算任务。每个模块在管理智能体行为、决策流程、记忆检索及网络交互方面各司其职，确保仿真流程的无缝衔接与高效运行。

4 模块实现

Casevo 通过设计并整合四个模块，实现了从智能体的自主决策到复杂社交网络的全方位仿真能力，为广泛复杂社会现象的研究与探索提供了高效且灵活的平台。

4.1 模型模块

模型模块用于构建模拟运行的环境，负责场景配置、事件调度、交互通道、提示模板及模型接口设置。此外，该模块还负责状态日志记录，确保在模拟过程中所有变更与操作均被完整记录。

- 场景设置：Casevo 读取用户提供的配置参数，从而定义仿真的初始环境。这包括确定智能体的数量、其角色以及仿真所需的其他关键参数。
- 事件调度：该模型模块负责控制仿真中事件的调度，确保智能体间的交互及时发生，并遵循基于轮次更新的机制所定义的逻辑流程。
- 交互通道：智能体通过定义明确的交互通道进行通信，从而能够共享信息、相互影响，并据此调整其行为。这些通道模拟了现实世界中的通信路径，例如社交媒体、公共辩论或直接对话。

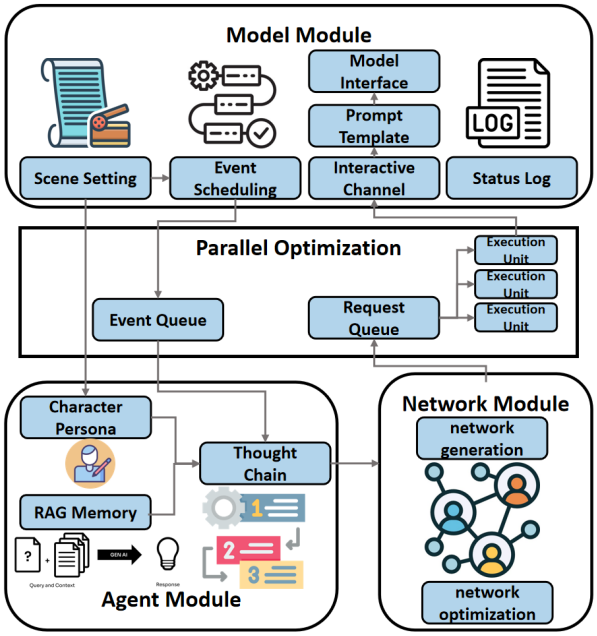


Figure 2: The architecture of the system.

- LLM Interfaces: The integration of LLMs allows agents to utilize multiple models and prompt templates for generating context-aware responses. These templates ensure consistency in agent communication, while the model interfaces support the interaction between the simulation environment and the LLMs, facilitating intelligent decision-making.
- State Logging: Throughout the simulation, the Model Module logs all changes in agent states, including decisions, interactions, and environmental changes. This data is crucial for post-simulation analysis, providing insights into how agents evolve and interact over time.

4.2 Agent Module

The Agent Module is responsible for defining the behavior of each agent, managing their memory, and enabling multi-step reasoning through the CoT mechanism. The agents in the simulation are highly customizable, with individual attributes, beliefs, and goals.

- Role Definition: Each agent is assigned a role, which governs its decision-making process and behavior patterns. These roles are often based on real-world entities, such as voters in an election or users in a social network, ensuring that agents exhibit realistic behaviors aligned with their defined attributes.
- RAG Memory: The Retrieval-Augmented Generation (RAG) memory system allows agents to recall past interactions and decisions, enabling more nuanced and context-aware behavior generation. This memory mechanism makes it possible for agents to retain information across multiple rounds of interaction, simulating human-like memory and long-term decision-making.

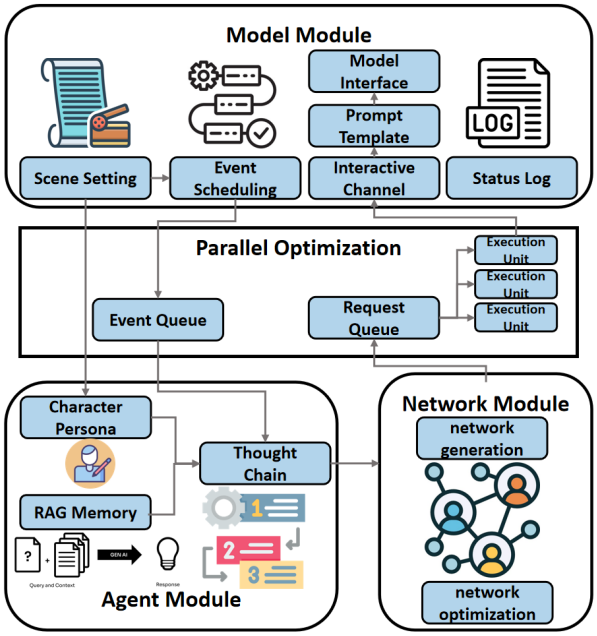


图 2：系统架构。

- LLM 接口：LLM 的集成使智能体能够利用多种模型及提示模板来生成具备上下文感知能力的响应。这些模板确保了智能体通信的一致性，而模型接口则支撑了仿真环境与 LLM 之间的交互，从而助力智能决策。
- 状态日志记录：在整个仿真过程中，模型模块会记录智能体状态的所有变化，包括决策、交互行为及环境变化。这些数据对于仿真后分析至关重要，可揭示智能体随时间推移如何演化及如何进行交互。

4.2 代理模块

代理模块负责定义每个代理的行为、管理其内存，并通过 CoT 机制实现多步推理。仿真中的代理具有高度可定制性，可拥有各自的属性、信念与目标。

- 角色定义：每个智能体都被分配一个角色，该角色决定了其决策过程与行为模式。这些角色通常基于现实世界中的实体而设定——例如选举中的选民或社交网络中的用户——从而确保智能体展现出与其既定属性相符的逼真行为。
- RAG 记忆系统：检索增强生成（RAG）记忆系统使智能体能够回顾既往交互记录与决策结果，从而生成更具细微差别且具备上下文感知能力的行为。该记忆机制使智能体能够在多轮交互中持续保留信息，模拟人类般的记忆能力与长期决策能力。

- CoT: The Mechanism provides agents with the ability to engage in multi-step reasoning. This mechanism allows agents to think through complex decisions, considering multiple factors before concluding. It is crucial for simulating strategic behaviors, such as planning, negotiation, and coalition-building.

4.3 Parallel Optimization Module

As the complexity of the simulation scale increases, the efficient computation becomes more and more urgent. The Parallel Optimization Module addresses this challenge by managing event and request queues, ensuring that agent actions are processed efficiently.

- Event Queue: The event queue manages all scheduled events within the simulation. Events can be triggered externally, such as through media broadcasts or social events, or internally, through agent interactions. The system ensures that events are processed in an organized manner, maintaining the flow of the simulation.
- Request Queue and Execution Units: When agents submit behavior requests, these are placed in the request queue, which is then processed by the system's execution units. By distributing the workload across multiple execution units, the system can handle parallel agent behaviors, ensuring that even large-scale simulations are processed on time.

4.4 Network Module

The Network Module is responsible for creating and managing the network structure within which agents operate. This module handles both the generation of the network at the start of the simulation and the dynamic adjustments that occur based on agent interactions.

- Network Generation: Casevo can generate social network structures based on the configuration parameters provided by users. These networks can take on different forms, such as small-world networks or random networks, depending on the type of social system being modeled.
- Network Optimization: Throughout the simulation, the network structure is dynamically adjusted based on agent interactions. For example, if two agents engage in frequent communication, the strength of their connection in the network may increase, reflecting their growing influence on each other. This dynamic adjustment ensures that the network structure evolves in a manner that mirrors real-world social networks.

4.5 Key Features

The system architecture of Casevo provides a robust foundation for multi-agent simulation, enabling the flexibility to meet diverse scenario requirements. Building on this architecture, a range of features further enhances its applicability and efficiency in simulating complex social behaviors.

a. Complex Network Support: Casevo provides powerful network generation and optimization capabilities, establishing flexible network structures for agent interactions. This structure not only simulates individual interactions within social networks but also dynamically adjusts agent relationships, reflecting the evolving characteristics of real-world social networks. This allows users to simulate diverse social systems, such as information dissemination in social media or social influence models.

- CoT：该机制赋予智能体进行多步推理的能力。它使智能体能够在做出复杂决策前，综合考虑多种因素，从而进行系统性思考。这对于模拟战略规划、谈判及联盟构建等策略性行为至关重要。

4.3 并行优化模块

随着仿真规模的日益复杂，高效计算的需求也愈发迫切。并行优化模块通过管理事件与请求队列，确保智能体动作得以高效处理，从而应对这一挑战。

- 事件队列：事件队列负责管理仿真环境中所有已预约的事件。事件可通过外部方式触发（例如媒体广播或社交活动），也可通过智能体间的交互从内部触发。该系统确保事件以有序方式进行处理，从而维持仿真的流程。
- 请求队列与执行单元：当智能体提交行为请求时，这些请求将被放入请求队列中，随后由系统的执行单元进行处理。通过将工作负载分布于多个执行单元，系统可并行处理智能体的行为请求，从而确保即使面对大规模仿真任务也能及时完成处理。

4.4 网络模块

网络模块负责创建并管理智能体运行所依赖的网络结构。该模块既在仿真开始时生成网络，也根据智能体间的交互动态调整网络结构。

- 网络生成：Casevo 可根据用户提供的配置参数生成社交网络结构。这些网络可呈现多种形态——例如小世界网络或随机网络——具体取决于所模拟社交系统的类型。
- 网络优化：在整个模拟过程中，网络结构会根据智能体间的交互关系进行动态调整。例如，若两个智能体频繁进行通信，它们在网络中的连接强度可能会增加，这反映了二者对彼此影响的增强。这种动态调整机制确保网络结构的演化过程能够真实模拟现实世界中的社交网络特征。

4.5 主要特性

Casevo 的系统架构为多智能体仿真提供了稳健的基础，使其具备灵活应对多样化场景需求的能力。基于该架构，一系列功能特性进一步提升了其在模拟复杂社会行为时的适用性与效率。

a. 复杂网络支持：Casevo 提供强大的网络生成与优化功能，可为智能体交互构建灵活的网络结构。该结构不仅能模拟社交网络中的个体交互，还可动态调整智能体间的关系，从而反映真实社交网络的动态演变特征。这使得用户能够模拟多样化的社交系统，例如社交媒体中的信息传播或社交影响力模型。

b. RAG Memory Mechanism: The system introduces a Retrieval-Augmented Generation (RAG) memory module, enabling agents to store and recall past events, decisions, and interaction histories. This memory mechanism allows agents to not only make decisions based on the current environment in each round but also adjust their actions based on historical data. This feature enables agents to simulate human-like long-term strategic thinking and memory-dependent decision-making.

c. Round-Based Updates: The system follows Mesa's round-based update model. In each round, agents simultaneously update their states and execute actions. This ordered round progression ensures the efficient handling of complex event scheduling and behavior synchronization across multiple agents.

d. Parallel Optimization: To improve simulation efficiency, Casevo incorporates parallel processing techniques into its execution engine. Through parallel optimization, the system can handle multiple agents' behaviors and decisions simultaneously, improving the overall simulation's performance. This feature makes Casevo particularly suitable for large-scale social system simulations, such as modeling voter behavior in elections and public opinion dissemination.

Through these enhancements, Casevo maintains the flexibility of the Mesa framework while enhancing its ability to handle complex semantic scenarios and large-scale network interactions. It stands as a powerful and versatile tool for multi-agent simulations. Later we will describe in more detail how Casevo achieves these features through a modular design and the specific functions and interactions of each module.

5 Scenario Example

One of the U.S. 2020 midterm election TV debates is utilized as an example to demonstrate the Casevo framework.

5.1 Scenario Settings

This election simulation is based on the 2020 US presidential election and simulates one of the TV public debates between candidates Trump and Biden. Agents play the role of voters to watch the debate, generate ideas or viewpoints based on their personas, and then conduct simulated voting. In addition, agents can interact and share their thoughts with each other, simulating the exchange of opinions and interactions between voters in reality.

To simulate the social interaction between voters, the social network structure is a small-world network with 101 nodes, as shown in Figure 3. Each node represents an agent or a voter. The edge between two nodes represents the friendship between the corresponding agents. The network allows agents to interact with each other, share their opinions, and adjust their voting preferences based on the content of the exchanges.

5.2 Agent Configuration

In the election simulation, the configuration of agents is crucial to ensuring that each simulated voter accurately reflects the complex psychology and behavior patterns of real voters in both behavior and decision-making. Each agent represents an individual who evaluates the candidates in the debate

b. RAG 记忆机制：本系统引入检索增强生成（RAG）记忆模块，使智能体能够存储并检索既往事件、决策及交互历史。该记忆机制不仅使智能体能在每轮交互中基于当前环境做出决策，还能根据历史数据调整其行为策略。此特性使智能体具备类人化的长期战略思维能力，以及依赖记忆的决策能力。

c. 基于轮次的更新机制：本系统采用 Mesa 框架中的基于轮次的更新模型。在每一轮迭代中，智能体同时更新其状态并执行动作。这种有序的轮次推进机制，可高效处理多智能体间的复杂事件调度与行为同步问题。

d. 并行优化：为提升仿真效率，Casevo 在其执行引擎中集成了并行处理技术。通过并行优化，该系统可同步处理多个智能体的行为与决策，从而提升整体仿真性能。该特性使 Casevo 特别适用于大规模社会系统仿真，例如选举中选民行为及公众舆论传播的建模。

通过这些改进，Casevo在保留Mesa框架灵活性的同时，显著增强了其处理复杂语义场景及大规模网络交互的能力。它已成为多智能体模拟领域一款强大且多功能的工具。下文我们将更详细地阐述Casevo如何通过模块化设计以及各模块的具体功能与交互机制来实现这些特性。

5 场景示例

本文以 2020 年美国中期选举中的电视辩论之一为例，用以演示 Casevo 框架。

5.1 场景设置

本次选举模拟基于2020年美国总统大选，再现了特朗普与拜登候选人之间的一场电视公开辩论场景。智能体扮演选民角色，观看辩论，根据其角色特征生成观点或见解，随后进行模拟投票。此外，智能体还可相互互动并分享观点，模拟现实中选民之间的意见交流与互动过程。

为模拟选民之间的社会互动，本研究采用一个包含 101 个节点的小世界社交网络结构，如图3所示。每个节点代表一个智能体或一名选民，两个节点之间的边则表示相应智能体之间的友谊关系。该网络允许智能体之间进行交互、分享观点，并基于交流内容调整其投票偏好。

5.2 代理配置

在选举模拟中，智能体的配置对于确保每个模拟选民在行为与决策层面均准确反映真实选民的复杂心理及行为模式至关重要。每个智能体代表一名个体选民，该个体选民根据候选人的特质及关

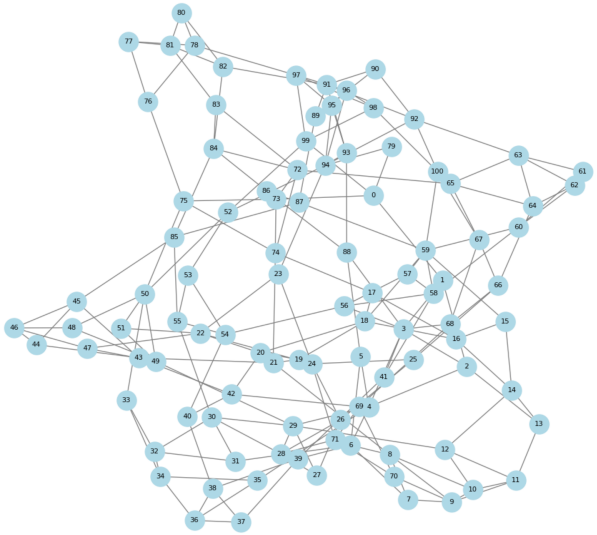


Figure 3: The structure of the network between voters.

based on their characteristics and areas of concern, making voting decisions based on the evaluation results. To describe an individual voter, the main parameters are as follows:

a. General description: The basic information of voters, including age, ethnicity, place of residence, etc.

b. Voter Characteristics: Includes background and topics of interest. Each voter agent has different areas of concern that shape their views on various national issues. For instance, some voters may advocate reducing government intervention in society, while others may insist on increasing it.

In order to simulate the real voter situation of the US presidential election, agents are configured with a study on the political typology of American voters [23]. It offers a nuanced classification beyond the simple Democratic-Republican division, identifying distinct groups such as steadfast conservatives, stable liberals, and young outsiders, each with unique political behaviors and attitudes. The agents in the experiment are categorized into multiple types based on the main characteristics of party voters in the political typology, with each type representing the typical characteristics of certain voter groups in reality, as shown in Table 1. This configuration allows the simulation to encompass voters from different age groups, social backgrounds, and political standings, and to model dynamic changes during the election process through these diverse settings.

Following the pre-defined agent configuration, the system generates voter agents that meet the role characteristics and quantity requirements through a rigorous initialization process. Table 2 is an example of agent profiles.

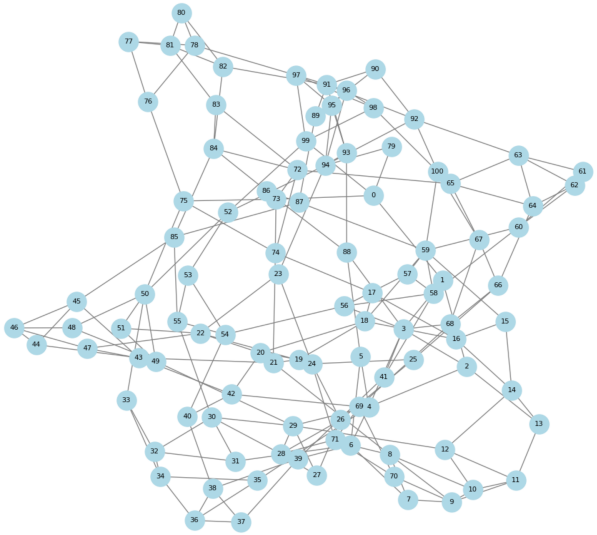


图 3：选民间网络的结构。

注领域对其进行评估，并基于评估结果做出投票决策。为刻画单个选民的特征，其主要参数如下：

a. 概况：选民的基本信息，包括年龄、民族、居住地等。

b. 选民特征：包括背景信息及关注议题。每位选民代理人的关注领域各不相同，这些领域塑造了他们对各类国家议题的看法。例如，部分选民可能主张减少政府对社会的干预，而另一些选民则坚持应增加此类干预。

为模拟美国总统大选中的真实选民情境，本研究基于美国选民政治类型学[23]配置了仿真主体。该类型学提供了超越简单“民主党-共和党”二分法的精细化分类，识别出诸如坚定的保守派、稳定的自由派以及年轻边缘群体等 distinct 选民群体，每个群体均具有独特的政治行为与态度。实验中使用的仿真主体根据政治类型学中政党选民的主要特征划分为若干类型，每种类型均代表现实中特定选民群体的典型特征，如表1所示。此种配置使仿真系统能够涵盖来自不同年龄层、社会背景及政治立场的选民，并借助这些多样化设定来模拟选举进程中的动态变化。

根据预设的代理配置，系统通过严格的初始化流程，生成符合角色特性与数量要求的投票代理。表2展示了代理配置示例。

Table 1: Voter Settings

Category	General Characteristics
Conservative White Male	55 years old, rural community, conservative
Financially Stable Educated White Male	Over 50, financially stable, educated white male
Low-Income, Low-Education White Female	Rural, low-income, low-education white female
Young, Diverse Group	Under 50, young, diverse
Low-Income White Female	Low-income, low education level white female
Financially Struggling Young Liberal Voter	Financially struggling, young, liberal
Elderly Religious Female	Elderly, deeply religious, low education level
High-Income, Highly Educated White	High-income family, highly educated white
Highly Educated Liberal Young White	Highly educated, group-oriented, young liberal white

Table 2: Profile Example

Name: Alex Thompson
Age: 55 years
Background Description: A 55-year-old conservative white male living in a rural community.
Topics of Interest: As a devout Christian, he believes religion is essential, supports a greater role for religion in public life, and feels government policies should uphold religious values. He actively participates in government and public affairs, advocating for reduced government intervention in society. On issues such as abortion and same-sex marriage, he holds conservative views, believing that the legalization of same-sex marriage harms the nation's interests. He firmly believes that being white does not confer any advantage based on race and that significant discrimination exists against white people in contemporary society. On international affairs, he believes that a strong national military is essential and supports military expansion, viewing military strength rather than diplomacy as the best means to ensure peace.

5.3 Simulation Steps

Since the presidential debate covers 6 main issues, the simulation is divided into 6 rounds. The main steps for agents in each round include: Watching and Evaluating, Discussing and Interacting, Reflecting and Voting.

5.3.1 Watching and Evaluating

At the beginning of each round, candidates broadcast their policy ideas and campaign concepts to voters through the TV debate. At this stage, all agents receive the corresponding transcripts of the debate simultaneously. Each voter processes this information according to his or her inclinations and personality traits to form an initial impression of the candidate as follows.

a. Parse the candidates' debate speech text and extract the information so that agents can analyze and assess the candidates' viewpoints.

表 1：投票设置

类别	一般特性
保守的白人男性	55岁，来自农村社区，性格保守
经济状况稳定、受过良好教育的白人男性	50岁以上、经济状况稳定、受过教育的白人男性
低收入、低学历的白人女性	农村、低收入、低教育水平的白人女性
年轻、多元化的团队	50岁以下，年轻且多元
低收入白人女性	低收入、低教育水平的白人女性
经济困难的年轻自由派选民	经济拮据、年轻、思想自由
年长女性信徒	老年人、虔诚的宗教信徒、教育水平较低者
高收入、高学历白人	高收入家庭、高学历白人
受过良好教育、思想开放的年轻白人	受过高等教育、注重团队协作、年轻且思想开放的白人

表 2：配置文件示例

姓名：Alex Thompson
年龄：55 岁

背景描述：一名55岁、性格保守的白人男性，居住于农村社区。

关注议题：作为一名虔诚的基督徒，他坚信宗教至关重要，支持宗教在公共生活中发挥更大作用，并认为政府政策应秉持宗教价值观。他积极参与政府及公共事务，主张减少政府对社会的干预。在堕胎与同性婚姻等议题上，他持有保守观点，认为同性婚姻合法化有损国家利益。他坚信，身为白人并不会基于种族身份获得任何优势，且当代社会中白人仍面临严重的歧视。在国际事务方面，他认为强大的国家军事力量至关重要，支持军事扩张，并主张以军事实力而非外交手段作为保障和平的最佳途径。

5.3 仿真步骤

由于总统辩论涵盖六大核心议题，本次模拟实验共分为六轮进行。每轮实验中，智能体的主要流程包括：观察与评估、讨论与互动、反思与投票。

5.3.1 观察与评估

每轮辩论开始时，候选人通过电视辩论向选民阐述其政策主张与竞选理念。在此阶段，所有参与者可同时获取相应辩论的文字记录。每位选民根据自身倾向与个性特征处理这些信息，进而对候选人形成初步印象，具体如下。

a. 解析候选人的辩论演讲稿文本，并提取相关信息，以便智能体能够分析并评估候选人的观点。

b. Based on the agents' interests and the content of the candidates' speeches, the agents evaluate the candidates.

c. Based on the results of the evaluation, the agents determine their level of support for each candidate.

At this point, agents have not yet made a final voting decision. They are only forming a tentative opinion about candidates. To demonstrate the information processing and decision-making process of the voter agent in the election simulation, Table 3 shows an example of the detailed content output generated by an agent.

Table 3: Example of the Watch and Evaluate Stage

"ts": 0,
"owner": "agent_60",
"type": "listen",
"item": {
"source": "public",
"content": "2020 U.S. Presidential Election .....",
"opinion": "
1. Fighting the new crown epidemic: supports Trump's anti-epidemic measures, including closing the border, promoting the use of masks, and accelerating vaccine research and development, but agrees with Biden on the need for increased testing and national standards for epidemic prevention.
2. Economy and jobs: endorses Trump's economic policies, especially tax cuts and measures to promote energy independence, while also supporting Biden's advocacy for more economic assistance for those affected by the epidemic.
3. Healthcare: Supports Trump's push for healthcare reform, including lowering drug prices and promoting Medicaid cooperation across state lines, but also endorses Biden's proposal to expand healthcare coverage.
4. Race relations: supports Trump's efforts on criminal justice reform and police department reform, while agreeing with Biden's response to systemic racism that police reform needs to go further.
5. Environment and energy: supports Trump's energy independence policies and environmental achievements, such as withdrawing from the Paris climate agreement, but agrees with Biden on the need to develop clean energy and reduce greenhouse gas emissions.
6. Foreign policy: recognizing Trump's achievements in diplomacy, such as diplomatic relations with North Korea and the Middle East peace process, but at the same time endorsing Biden's advocacy for greater global cooperation to address international challenges."
}

5.3.2 Discussing and Interacting

Agents interact and exchange opinions with their neighbors (other agents in their social circle) through social networks. Each agent will not only share their opinions about the debate content, but will also receive ideas and feedback from others.

As shown in Table 4, in the first round Agent No.60 and Agent No.62 generate an exchange in which Agent No.60 expresses his or her views on the six main topics of the debate to Agent No.62,

b. 代理人根据自身利益及候选人的演讲内容，对候选人进行评估。

c. 根据评估结果，各代理体为每个候选对象确定其支持程度。

在此阶段，代理尚未做出最终的投票决策，仅对候选人形成初步判断。为展示选举模拟中投票代理的信息处理与决策流程，表3展示了该代理生成的详细输出内容示例。

表 3：观察与评估阶段示例

“ts”: 0，“owner”:
“agent_60”，“type”:
“listen”，“item”: {
“来源”:“公开”，“内容”:“2020年美国总统大
选”，“观点”:“”
1. 在应对新冠疫情方面，他支持特朗普的抗疫措施——包括关闭边境、推广佩戴口罩以及加速疫苗研发——但同意拜登关于加强检测力度及制定全国防疫标准的观点。
2. 经济与就业方面：支持特朗普的经济政策，尤其是减税措施及促进能源独立的举措；同时支持拜登关于为受疫情影响人群提供更多经济援助的倡议。
3. 医疗保健领域：支持特朗普推动的医疗改革举措——包括降低药品价格及促进跨州 Medicaid 合作——同时也支持拜登提出的扩大医疗覆盖范围的提案。
4. 种族关系：支持特朗普在刑事司法改革及警察部门改革方面所做的努力，同时认同拜登针对系统性种族主义提出的回应——即警察改革需进一步深化。
5. 环境与能源：支持特朗普的能源独立政策及环境领域的成就——例如退出《巴黎气候协定》；但在发展清洁能源与减少温室气体排放的必要性问题上，与拜登持相同立场。
6. 外交政策：在认可特朗普在外交领域取得的成就——例如与朝鲜建立的外交关系及中东和平进程——的同时，也支持拜登倡导加强全球合作以应对国际挑战。
}

5.3.2 讨论与互动

智能体通过社交网络与其邻近智能体（社交圈中的其他智能体）进行交互并交换观点。每个智能体不仅会分享其对辩论内容的看法，还会从其他智能体处获取新思路与反馈。

如表4所示，在第一轮对话中，智能体 60 与智能体 62 进行了一次交换，其中智能体 60 向智能体 62 阐述了其对辩论中六个主要议题的看法。

Table 4: Examples of Opinion Expressions in Voter Interaction

Epidemic response: supports Trump's anti-epidemic measures (e.g., closing the border, vaccine development) but agrees with Biden's point about increased testing and national standards for prevention. **ECONOMY & JOBS:** Endorses Trump's tax cuts and promotion of energy independence, but agrees with Biden's assertion of the need for financial assistance for those affected by the outbreak. **Healthcare:** supports Trump's health care reform, but believes it needs to be available to more people, endorses some of Biden's proposals. **Race relations:** supports Trump's efforts on justice reform, while agreeing with Biden's stance on tackling systemic racism. **ENVIRONMENT & ENERGY:** Supports Trump's energy independence policy while endorsing Biden's proposal to develop clean energy. **Foreign policy:** supports Trump's foreign policy accomplishments, but sees the need for greater global cooperation to address international challenges, partially agreeing with Biden.

and the content of the exchange is recorded in the voter's memory, and then that voter revisits the two candidates in light of the content of the discussion, generating a new viewpoint, as detailed in Table 5.

Table 5: Example of post-interaction reflection


```
"agree": {
  "Trump": {
    "Identity": 0.6,
    "Overall view": "Generally supportive, especially on tax cuts, deregulation, and energy independence, but has concerns about its health care reform and neglect of climate change."
  },
  "Biden": {
    "Identity": 0.4,
    "Overall View": "There is consensus on issues such as economic aid, police reform, and clean energy, but disagreement on immigration policy, which is partially in favor of its policies."
  }
}
```


5.3.3 Reflecting and Voting

At the end of each round, agents entered a reflective phase, as shown in Table 6. At this point, each agent retrieves and reflects on its short-term memory based on the previous information and the discussion content, and further adjusts its views by combining them with past historical experiences (e.g., previous candidate performance or past policies). Through this built-in reflective mechanism, agents not only rely on the current discussion, but also synthesize their experiences and memories over time. This stage helps agents make more solid and rational decisions, especially when they are confronted with complex social issues or policy controversies.

表 4：选民互动中观点表达的示例

疫情应对：支持特朗普的抗疫措施（如关闭边境、推进疫苗研发），但认同拜登关于加强检测及制定全国预防标准的观点。**经济与就业：**支持特朗普的减税政策及推动能源独立的举措，但认同拜登关于需为疫情受影响人群提供财政援助的论断。

医疗保健：支持特朗普的医改方案，但认为该方案应惠及更多民众；同时支持拜登提出的部分提案。

种族关系：支持特朗普推动的司法改革举措，同时赞同拜登应对系统性种族主义的立场。**环境与能源：**支持特朗普的能源独立政策，并赞同拜登发展清洁能源的提案。

外交政策：认可特朗普的外交政策成就，但认为为应对国际挑战，需要加强全球合作，并在部分问题上与拜登持相同立场。

讨论内容将被记录在投票者的记忆中，随后该投票者根据讨论内容重新审视两位候选人，进而形成新的观点，具体如表5所示。

表 5：互动后反思示例


```
"agree": {
  "Trump": {
    "Identity": 0.6,
    "Overall view": "Generally supportive, especially on tax cuts, deregulation, and energy independence, but has concerns about its health care reform and neglect of climate change."
  },
  "Biden": {
    "Identity": 0.4,
    "Overall View": "There is consensus on issues such as economic aid, police reform, and clean energy, but disagreement on immigration policy, which is partially in favor of its policies."
  }
}
```


5.3.3 反思与投票

每轮结束时，智能体都会进入反思阶段，如表6所示。在此阶段，每个智能体会根据先前信息及讨论内容检索并反思其短期记忆，进而将其与既往历史经验（如既往候选者表现或既往政策）相结合，进一步调整自身观点。通过这一内置反思机制，智能体不仅依托当前讨论，还能整合其随时间推移积累的经验与记忆。这一阶段有助于智能体做出更为稳健且合理的决策，尤其在面对复杂社会问题或政策争议时尤为有效。

Table 6: Example of a reflection phase


```
"item": {
  "ori_opinion": null,
  "new_opinion": "1. TRUMP: I support President Trump's positions on many key issues, including religious liberty, reducing the size of government, and the importance of national defense. At the same time, I agree with the steps he has taken on economic policy, health care, race relations, environment and energy, and foreign policy. Despite my reservations about some of his statements and actions, I still believe he is the candidate who better represents my views.
Biden: While I have reservations about Vice President Biden's views on some issues, such as his attitude toward illegal immigration and his views on public assistance, I also believe that he has put forward sound programs in other areas such as health care, race relations and environmental policy. However, I am concerned about his tendency to support same-sex marriage and expand government intervention. Overall, I think Vice President Biden differs significantly from my views in some areas, but in others there is some agreement."
}
```


After reflection, agents make their final voting choices based on the results of the reflection. Each agent's voting decision is directly influenced by the previous phases, including the effectiveness of the candidate's speech, the flow of opinions in the social network, and their own reflective process. The system records each voter's voting choice, providing data for status updates in subsequent rounds and for the final results tally.

Table 7: Example of a Voting Stage


```
"ts": 0,
"owner": "agent_60",
"type": "vote",
"item": {
  "Trump": 0.8,
  "Biden": -0.5
},
"owner_type": 0
```


The election example demonstrates how Casevo's modular design can be useful in the simulation of complex social phenomena. By coordinating agent behavior, dynamically adapting social networks, and employing advanced memory and reasoning mechanisms, the system provides a powerful tool for exploring the election process in depth, demonstrating how individual decisions can affect collective outcomes.

5.4 Evaluation

Figure 4(a) shows the changes in the agents' attitudes toward the candidates and their voting behaviors in each round. It can help researchers understand the dynamic evolution process of voters' opinions.

As shown in Figure 4(a), Biden's approval rating is relatively high in the initial stages and

表 6：反射相示例

“项目”: {
“ori_opinion”: null,
“new_opinion”: “1. 特朗普：我支持特朗普总统在诸多关键议题上的立场，包括宗教自由、缩减政府规模以及国防的重要性。与此同时，我也赞同他在经济政策、医疗保健、种族关系、环境与能源以及外交政策方面所采取的举措。尽管我对他部分言论及行为存有保留意见，但我仍认为他是最能代表我观点的候选人。”
拜登：尽管我对副总统拜登在某些问题上的观点——例如他对非法移民的态度以及对公共援助的看法——持有保留意见，但我同样认为，他在医疗保健、种族关系及环境政策等其他领域提出了切实可行的方案。然而，我对他支持同性婚姻及扩大政府干预的倾向感到担忧。总体而言，我认为拜登副总统在某些方面与我存在显著分歧，但在其他方面我们仍具一定共识。
}

经过反思后，各智能体根据反思结果做出最终投票选择。每个智能体的投票决策均直接受前期阶段的影响，包括候选人的演讲效果、社交网络中的观点流动情况以及自身反思过程。该系统记录每位投票者的投票选择，为后续轮次的状态更新及最终结果统计提供数据支持。

表 7：投票阶段的示例

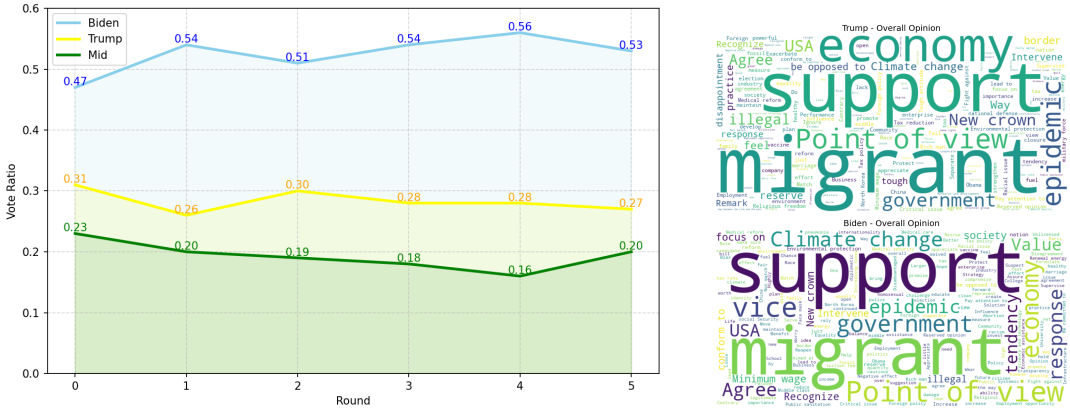
“ts”: 0，“owner”:
“agent_60”，“type”:
“vote”，“item”: {
“特朗普”: 0.8,
“拜登”: -
0.5 }，“所有者
类型”: 0

这一选举案例展示了 Casevo 的模块化设计如何在复杂社会现象的模拟中发挥重要作用。通过协调智能体行为、动态调整社交网络结构，以及采用先进的记忆与推理机制，该系统为深入探究选举过程提供了强大工具，进而揭示个体决策如何影响集体结果。

5.4 评估

图4(a)展示了各轮次中参与者对候选人的态度及其投票行为的变化情况。该图有助于研究人员理解选民观点的动态演变过程。

如图4(a)所示，拜登的支持率在选举初期相对较高，并随着选举进程的推进稳步上升，最终稳



(a) Voting Result

(b) Word cloud of voter opinions

Figure 4: Simulation Result

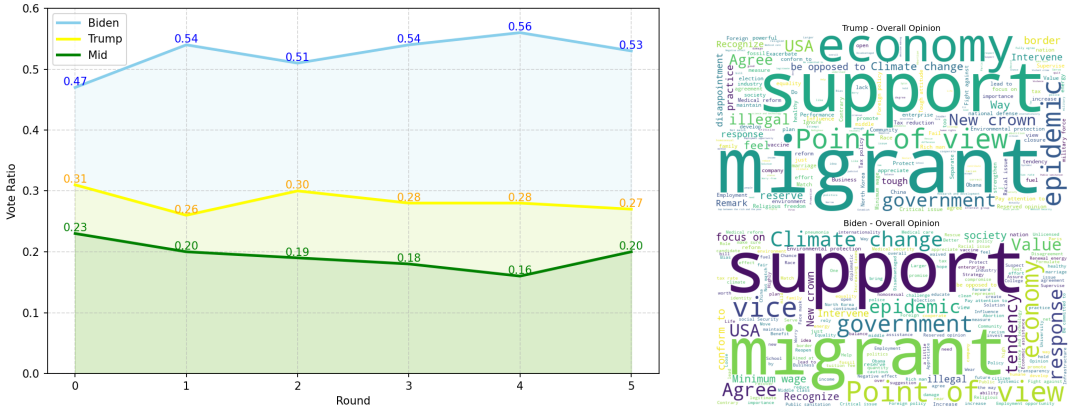
steadily increases as the election progresses, eventually stabilizing at a high level. Trump's support, on the other hand, shows a downward trend at the beginning of the election, and despite a rebound in some rounds, the overall support fails to pick up significantly. The proportion of neutral voters decreases, suggesting that more and more neutral voters begin to favor either Biden or Trump as the election progressed, but more neutral voters turns to Biden.

This result suggests that Biden's policies and speeches won over more voters during the simulation, especially in the mid-to-late-debate period, when Biden tilts neutral voters towards his camp through his sustained policy advocacy and communication. Trump, on the other hand, fails to effectively expand his base of supporters, and his support fluctuates and eventually declines slightly.

As shown in Figure 4(b), opinions about Biden are dominated by positive words such as "support" and "approve", reflecting Biden's strong support among voters. Voters highly approve of Biden's stance on several policy issues, especially on issues such as the response to the new crown epidemic, environmental and energy policies, and race relations, where Biden's policy ideas are consistent with voters' expectations and have gained widespread support.

By contrast, Opinions about Trump are almost evenly split between positive and negative terms. This suggests that voters are sharply divided on Trump's positions, particularly on key issues such as the response to the epidemic, race relations, and foreign policy, where voters are more antagonistic in their views. A portion of voters hold a favorable view of his policies, while another portion is strongly opposed to them. Overall, Trump's support in the simulation is relatively weak, and his governance strategy fails to win the approval of a wide range of voters.

The experimental results clearly show the distribution of voters' support for the two candidates and their dynamics through the graphs of changes in support and word cloud maps of voters' perceptions. Biden shows a stronger support tendency in the simulation, while Trump's support is more fluctuating, and these results provide an important data basis for the simulation study of election dynamics.



(a) 投票结果　　(b) 选民意见词云

图 4：仿真结果

定在较高水平；而特朗普的支持率则在选举初期呈现下降趋势，尽管在部分投票轮次中有所回升，但其总体支持率仍未能显著改善。中立选民的比例有所下降，这表明随着选举推进，越来越多的中立选民开始倾向于支持拜登或特朗普，但其中转向支持拜登的比例更高。

这一结果表明，在模拟推演过程中，拜登的政策与演讲吸引了更多选民支持，尤其是在辩论的中后期阶段——彼时拜登通过持续的政策倡导与沟通，成功将中立选民倾向其阵营。相比之下，特朗普则未能有效扩大其支持基础，其支持率呈现波动态势，并最终略有下降。

如图4(b)所示，对拜登的评价以“支持”、“赞同”等积极词汇为主，反映出拜登在选民中享有的广泛支持。选民对拜登在若干政策议题上的立场高度认可，尤其是在应对新冠疫情、环境与能源政策以及种族关系等问题上——拜登的政策主张与选民预期相符，并已获得广泛支持。

相比之下，对特朗普的评价在正面与负面之间几乎平分秋色。这表明选民在特朗普的立场问题上存在显著分歧，尤其是在应对疫情、种族关系及外交政策等关键议题上，选民的观点更为对立。部分选民对其政策持肯定态度，而另一部分选民则强烈反对这些政策。总体而言，特朗普在本次模拟选举中的支持率相对较低，其执政策略未能赢得广大选民的认可。

实验结果通过支持率变化曲线图及选民认知词云图，清晰展示了选民对两位候选人的支持分布及其动态变化。在模拟实验中，拜登的支持倾向更为显著，而特朗普的支持率则呈现较大波动；这些结果为选举动态模拟研究提供了重要的数据基础。

6 Discussion and Future Work

In the previous sections, we presented in detail the system architecture, module design, and specific applications of Casevo in election simulation. Casevo enables a high-quality social simulation and demonstrates its power in complex scenarios. Firstly, in terms of ease of use, the system allows users to easily configure agents and simulation scenarios, avoiding the tedious task of writing complex code. Users can generate complex simulation scenarios by simply entering the appropriate parameters according to their needs, greatly reducing the learning threshold and operational difficulty. Secondly, in terms of efficiency, Casevo is optimized for the parallel behavior of large-scale agents, especially in complex interaction and decision processing. By using parallelization techniques, the system can process a large number of tasks in a short time, greatly improving simulation efficiency. In addition, Casevo has excellent real-time performance, providing instant feedback on the dynamic behavior and decision results of the agent during the simulation process, allowing users to monitor and understand the progress of the simulation in real-time. Together, these features provide a powerful and efficient simulation tool, and the highly modular design of the Casevo architecture allows the system to be easily adapted to different application scenarios.

In future work, we consider introducing more transparent behavioral models or incorporating more logical reasoning mechanisms to enhance the interpretability of agent behavior. At the same time, we explore more effective data integration methods to ensure that the system can better reflect the dynamic processes in the real world; and try to use larger scale and more realistic social network data, as well as more detailed individual behavioral data, to further validate the effectiveness and universality. In addition, besides election simulation, the Casevo architecture can be applied to other simulation scenarios of social dynamics. For example, news dissemination, opinion generation, and market behavior. We plan to integrate more social science theories and behavioral models into Casevo through modular design, so as to enhance its performance capability in different application scenarios.

7 Conclusion

In this paper, we propose the Casevo framework, which can help implement a multi-agent system capable of modeling complex social dynamics by integrating LLMs as a decision center. We demonstrate the application of the Casevo in election simulation by creating a large number of voter agents to reproduce the information dissemination, interaction and decision-making behavior during the election process. The results show that the Casevo system is effective in capturing the complexity of voter behaviors and provides new perspectives and tools for election research. Casevo demonstrates strong flexibility and scalability in simulation applications. Future research directions include optimizing the performance of LLMs, improving the quality of generated content, and exploring more effective methods for real-world data integration. In addition, we will extend the application scenarios of Casevo to further validate its broad applicability in social science research.

References

[1] John H Miller and Scott E Page. *Complex adaptive systems: an introduction to computational models of social life: an introduction to computational models of social life*. Princeton university press, 2009.

6 讨论与未来工作

在之前章节中，我们已详细阐述了 Casevo 的系统架构、模块设计及其在选举模拟中的具体应用场景。Casevo 支持高质量社会模拟，并在复杂场景中展现出强大性能。首先，在易用性方面，该系统允许用户便捷地配置智能体与模拟场景，无需编写复杂代码即可完成配置，从而避免繁琐操作；用户仅需根据需求输入相应参数，即可生成复杂的模拟场景，显著降低学习门槛与操作难度。其次，在效率方面，Casevo 针对大规模智能体的并行运行进行了优化，尤其在复杂交互与决策处理场景中表现突出。通过采用并行化技术，系统可在短时间内处理大量任务，大幅提升模拟效率。此外，Casevo 具备卓越的实时性能，可在模拟过程中即时反馈智能体的动态行为与决策结果，使用户能够实时监测并理解模拟进程。这些特性共同构成了一个强大且高效的仿真工具，而 Casevo 架构的高度模块化设计则使该系统能够轻松适配不同的应用场景。

在未来的研究中，我们计划引入更透明的行为模型，或整合更多逻辑推理机制，以提升智能体行为的可解释性。与此同时，我们将探索更有效的数据整合方法，确保系统能更精准地反映现实世界的动态过程；并尝试采用更大规模、更具真实性的社交网络数据，以及更详尽的个人行为数据，以进一步验证该系统的有效性与普适性。此外，除选举模拟外，Casevo 架构还可应用于其他社会动态模拟场景，例如新闻传播、观点生成及市场行为模拟。我们计划通过模块化设计，将更多社会科学理论与行为模型整合至 Casevo 中，从而增强其在不同应用场景下的性能表现。

7 结论

本文提出 Casevo 框架，该框架通过将大语言模型（LLM）作为决策核心进行整合，助力构建能够建模复杂社会动态的多智能体系统。我们通过创建大量选民智能体，以复现选举过程中的信息传播、互动及决策行为，展示了 Casevo 在选举模拟中的应用效果。实验结果表明，Casevo 系统可有效捕捉选民行为的复杂性，为选举研究提供新的视角与工具。Casevo 在模拟应用中展现出卓越的灵活性与可扩展性。未来研究方向包括优化大语言模型的性能、提升生成内容的质量，以及探索更有效的现实世界数据整合方法。此外，我们还将拓展 Casevo 的应用场景，以进一步验证其在社会科学研究中的广泛适用性。

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